# Na Li


Incoming PhD Student in Language Science and Technology

The Hong Kong Polytechnic University | Starting September 2026

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## Research Profile

Incoming PhD student with interdisciplinary training in electronic engineering, psychology, and cognitive neuroscience. My research uses EEG, intracranial EEG (sEEG), and computational approaches to investigate neural representations and dynamics underlying language and cognition. My current interests focus on the neural mechanisms of natural language understanding, including neural encoding and decoding and computational models of language processing.

## Research Interests

Language neuroscience | Neural mechanisms of natural language understanding | Neural encoding and decoding | Computational cognitive neuroscience

## Education

**Incoming PhD Student, Department of Language Science and Technology** — Starting Sep. 2026
The Hong Kong Polytechnic University, Hong Kong; Supervisor: Prof. Shaonan Wang

**M.A. in Psychology** — Sep. 2022 - Jun. 2025
Southwest University, Chongqing, China

**B.Eng. in Electronic Science and Technology** — Sep. 2018 - Jun. 2022
Hangzhou Dianzi University, Hangzhou, China

## Publications and Research Outputs

**Peer-reviewed article**

- **Li, N.**, Yang, J., Long, C., & Lei, X. (2024). Test-retest reliability of EEG aperiodic components in resting and mental task states. *Brain Topography, 37*(6), 961-971. doi:10.1007/s10548-024-01067-x

**Manuscript under review**

- Wei, Y., **Li, N.**, Mao, R., & Long, C. (2025). Examining self-other distinction inefficiency and egocentric processing in lonely individuals. Manuscript under review.

**Patent application**

- Kong, X., Zhou, D., **Li, N.**, & Pu, Y. (2025). A method for selecting language function rehabilitation corpus based on semantic embedding and neural encoding. Chinese Patent Application No. CN202511371878.0. Pending.

## Research Experience

**Research Assistant** Sep. 2025 - Present

*Zhejiang University, Department of Psychology and Behavioral Sciences*

Supervisor: Prof. Xiangzhen Kong

- **Aperiodic activity in spatial navigation:** Investigate aperiodic components in sEEG recordings to characterize temporal neural dynamics during spatial navigation and path integration.
- **Semantic decoding for language rehabilitation:** Contributed to semantic-embedding and neural-encoding approaches for rehabilitation-corpus selection; led the drafting of a related Chinese patent application.

**Graduate Researcher** Jan. 2023 - Jun. 2025

*Southwest University, Faculty of Psychology*

Supervisor: Prof. Changquan Long; Co-supervisor: Prof. Xu Lei

- **Reliability of aperiodic EEG activity:** Constructed and analysed a multi-state EEG dataset; compared spectral-parameterisation methods across data durations and task states; published the findings as a first-author article in *Brain Topography*.
- **Temporal expectation and selective attention:** Independently designed and conducted an EEG study of temporal expectation and distractor suppression; analysed behavioural, ERP, and time-frequency measures for my master's thesis.
- **Self-other distinction:** Contributed to study design and EEG analysis in a visual perspective-taking study of loneliness and egocentric interference.
- **Multidimensional emotional music analysis:** Led multidimensional analysis, visualisation, and manuscript preparation for a study of emotional experience in music.

## Selected Awards

**Second Prize, Outstanding Graduate Scholarship** Nov. 2024

Southwest University

**Zhejiang Provincial Government Scholarship** Oct. 2020

Zhejiang Province, China

**Meritorious Winner, Mathematical Contest in Modeling (Top 8% worldwide)** Apr. 2020

COMAP MCM/ICM

## Methods and Technical Skills

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|---|---|
| **Neural data analysis** | EEG, sEEG, ERP, time-frequency analysis, spectral parameterisation |
| **Programming and statistics** | Python, MATLAB, R, JASP, SPSS |
| **Neuroimaging toolboxes** | MNE-Python, FieldTrip, Brainstorm, EEGLAB |
| **Experimental tools** | PsychoPy, E-Prime, OpenSesame, Experiment Builder |
| **Experimental methods** | EEG, eye-tracking, TMS |

## Languages

Mandarin Chinese (native) | English (IELTS 7.0)